

Disproof of Conjecture 00000009626

The three-symbol full shift

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Abstract

The unrestricted class of two-dimensional shifts of finite type contains the full shift on three symbols. Its spatial entropy is $\log 3$, contradicting the asserted upper bound $\log 2$. We give the complete counting argument and a Lean 4 formalization of the configurations, finite-type condition, pattern counts, real logarithms, and entropy limit.

The assertion addressed

Conjecture 00000009626 asserts, in both its English and Chinese versions, that the set of realizable spatial entropies of two-dimensional shifts of finite type (SFTs) is $[0, \log 2]$. Neither version restricts the alphabet to two symbols. We use the ordinary definition allowing any finite alphabet. The independent assertions about row-by-row approximation are unnecessary: the claimed entropy interval is already false.

An explicit SFT and its entropy

Let $A = \{0, 1, 2\}$ and let

$$X = A^{\mathbb{Z}^2}.$$

This is the full shift: every configuration is allowed. It is a shift of finite type defined by the empty finite collection of forbidden patterns. It is nonempty, closed, and invariant under every lattice translation.

For $n \geq 1$, write $Q_n = \{0, \dots, n-1\}^2$, and let $N_X(n)$ be the number of patterns on Q_n occurring in configurations in X . Every map $p : Q_n \rightarrow A$ occurs: extend it to $\mathbb{Z}^2$ by assigning the symbol 0 outside Q_n . Conversely every occurring pattern is such a map. Since $|Q_n| = n^2$, independent choices at all sites give exactly

$$N_X(n) = 3^{n^2}.$$

Using the usual square-box definition of spatial entropy (natural logarithms),

$$h(X) = \lim_{n \rightarrow \infty} \frac{\log N_X(n)}{|Q_n|} = \lim_{n \rightarrow \infty} \frac{\log(3^{n^2})}{n^2} = \log 3.$$

The normalized expression is identically $\log 3$ for every positive n , so there is no approximation or unproved limiting interchange in this step.

Proposition 1. *The entropy spectrum asserted in Conjecture 00000009626 is not $[0, \log 2]$.*

Proof. The SFT X constructed above has entropy $\log 3 > \log 2$, because the natural logarithm is strictly increasing on the positive real numbers. Thus its entropy lies outside the asserted interval. $\square$

Scope and formal verification

This counterexample uses three symbols and addresses the unrestricted statement that was submitted. Adding a binary-alphabet restriction would change that statement; the present example makes no claim to disprove an upper bound for binary subshifts. No classification theorem for SFT entropies is needed here. For background on the actual unrestricted spectrum, see Hochman and Meyerovitch [1].

The accompanying `lean/Main.lean` defines actual configurations $(\mathbb{Z} \times \mathbb{Z}) \rightarrow \text{Fin}(3)$, SFTs through finite forbidden square patterns at every translate, and occurring patterns through extension to an actual configuration. It proves an equivalence between all square patterns and occurring patterns of X , then computes their cardinality. `HasSpatialEntropy` is the convergence of the normalized real logarithmic counts along boxes of side $n + 1$; this harmless reindexing excludes the empty box. The main theorem `counterexample` provides an SFT, its proved entropy limit, and the strict violation of the proposed bound. `not_universal_entropy_bound` negates the bound explicitly.

The project pins Lean 4.19.0 and Mathlib v4.19.0 at commit

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

It contains no admitted theorem or added axiom. The supplemental Python script checks small finite examples only; the proof for every box size and the real limit are in Lean. Build logs, printed foundational axiom dependencies, file hashes, and a separate adversarial self-review are provided with the submission.

References

- [1] M. Hochman and T. Meyerovitch, *A characterization of the entropies of multidimensional shifts of finite type*, Annals of Mathematics 171 (2010), 2011–2038. [arXiv:math/0703206](https://arxiv.org/abs/math/0703206).